



Resources allowed: A handwritten A4 notes page double sided

Name: Solutions

Class: _____

Total Time: 55minutes

Total Mark/49

Non-Calculator Mark /13

Calculator Mark /36

<u>NON-CALCULATOR SECTION</u>

1) [4 marks]

Fill in the missing number in each of the following

a) $6 + (-8) = \underline{-2}$ ✓

b) $-4 - (-7) = \underline{3}$ ✓

c) $3 \times (-7) = \underline{-21}$ ✓

d) $-16 \div (-2) = \underline{8}$ ✓

2) [1, 1, 1, 2 = 5 marks]

Express the following in **simplest index form** using the index laws

a) $(4^3)^3 = 4^9$ ✓

b) $\frac{6^5}{6^2} = 6^3$ ✓

c) $2^3 \times 2^4 = 2^7$ ✓

d) $\frac{(5^2)^3 \times 2^6 \times 2^3}{5^2 \times 2^5} = \frac{5^6 \times 2^9}{5^2 \times 2^5} = 5^4 \times 2^4$ ✓

3) [1, 1, 1, 1 = 4 marks]

Evaluate the following expressions, writing your answers as **basic numerals (numbers)**:

a) $2^0 = 1$ ✓

b) $\frac{5^3}{5^3} = 1$ ✓

c) $3^2 + 3^0 = 9 + 1 = 10$ ✓

d) $2^5 = 32$ ✓



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 Name: Solutions Class: _____ Total Time: 50minutes Calculator Allowed Mark /36

<u>CALCULATOR ALLOWED SECTION</u>
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3) [3 marks]

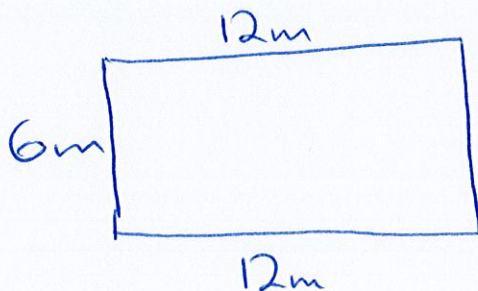
Complete the following conversions:

- a) $3\text{m} = \underline{3000}\text{mm}$ ✓
- b) $2\text{cm}^2 = \underline{200}\text{mm}^2$ ✓
- c) $5000\text{cm}^3 = \underline{0.005}\text{m}^3$ ✓

4) [1, 1 = 2 marks]

The perimeter of a rectangle is 36 m.

a) Draw a rectangle with this perimeter and mark in a possible length and width.



Any rectangle
with $P = 36\text{m}$

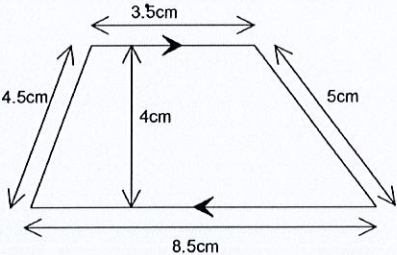
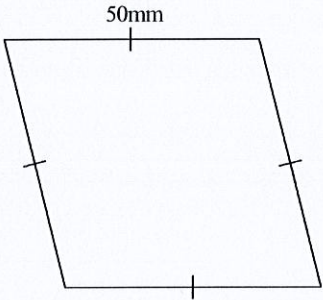
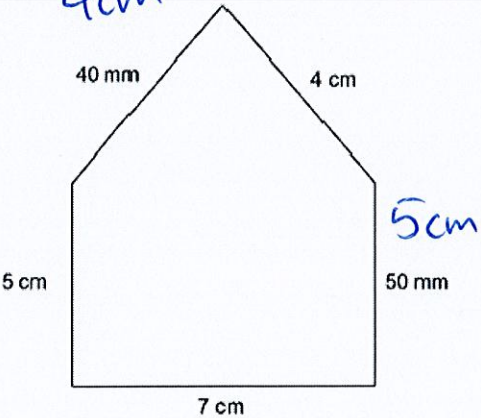
b) Find the area of your rectangle.

$$\begin{aligned}
 A &= L \times w \\
 &= 12 \times 6 \\
 &= 72\text{m}^2 \quad \checkmark
 \end{aligned}$$

Any correct area calculation
for above rectangle

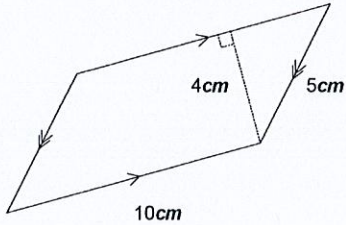
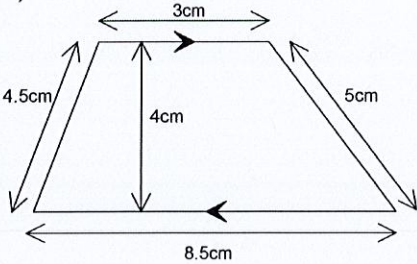
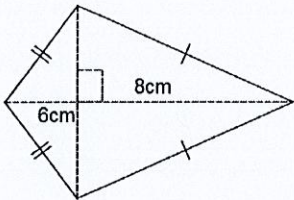
5) [1, 1, 2, + 1 mark for **all** units =5 marks]

Find the **perimeter** of the following shapes. Show your working clearly and giving your answers correct to two decimal places when necessary:

Shape	Working and Answer
a) 	$3.5 + 5 + 8.5 + 4.5$ $= 21.5 \text{ cm} \checkmark$
b) 	$50 + 50 + 50 + 50$ $= 200 \text{ mm} \checkmark$
c) 	$\checkmark$ (converts mm to cm correctly) $4 + 4 + 5 + 7 + 5$ $= 25 \text{ cm} \checkmark$

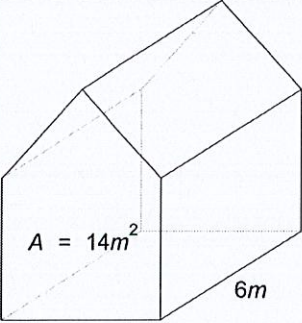
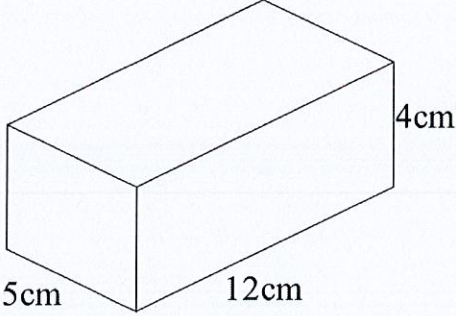
6) [1, 2, 1 = 4 marks]

Find the **area** of the following shapes. Show your working clearly and give your answers correct to two decimal places when necessary.

Shape	Working and Answer
a) 	$A = b \times h = 10 \times 4 = 40 \text{ cm}^2 \checkmark$
b) 	$A = \frac{1}{2} (a+b) \times h \checkmark$ $= \frac{1}{2} (3+8.5) \times 4$ $= 23 \text{ cm}^2 \checkmark$
c) 	$A = \frac{1}{2} xy = \frac{1}{2} \times 6 \times 8$ $= 24 \text{ cm}^2 \checkmark$

7) [2, 2 = 4 Marks]

Calculate the **volume** of this prism:

Shape	Working and Answer
	$V = A \times d \checkmark$ $= 14 \times 6$ $= 84 \text{ m}^3 \checkmark$
	$V = L \times w \times h \checkmark$ $= 240 \text{ cm}^3 \checkmark$

8) [2, 2 = 4 marks]

- a) Riley's tent makes a triangular prism when it is put up. He knows the tent is 1.2m wide and 1.5m high. He does not know what the volume of the tent is, so he measures the length as 180cm. What is the volume of the tent?

$$A = \frac{1}{2} \times b \times h \checkmark$$

$$= \frac{1}{2} \times 1.2 \times 1.5 = 0.9 \text{ m}^2$$

$$V = A \times L$$

$$= 0.9 \times 1.8$$

$$= 1.62 \text{ m}^3 \checkmark$$

- b) What would the length of the Ethan's tent be for the volume to be 2 m^3 ?

$$V = A \times L$$

$$2 = 0.9 \times L \checkmark$$

$$L = \frac{2}{0.9} = 2.2 \text{ m} \checkmark$$

2.2 m , any number of decimal places acceptable.

9) [1, 2, 2 = 5 marks]

Here are the results of a survey of cars passing the school.

Colour	Red	Blue	Yellow	Black	White
Number of cars	7	19	4	23	27

a) What is the experimental probability of a red car passing the school?

$$7 + 19 + 4 + 23 + 27 = 80$$

$$P(\text{red}) = \frac{7}{80}$$

b) Describe the complement of the event in part (a) and find its experimental probability of occurring.

A car that is not red ✓

$$= 1 - \frac{7}{80} = \frac{73}{80} \checkmark$$

c) If a student was to repeat this survey again at the same time next week, recording colours until 100 cars had passed, how many of them would she expect to be yellow?

$$\frac{4}{80} = \frac{5}{100} \checkmark \quad 5 \text{ yellow cars} \checkmark$$

10) [1, 1, 1 = 3 marks]

Write the complementary event for the following:

a) It will rain today

It will not rain today ✓

b) Rolling a 6-sided die and getting a 5

Rolling a 6-sided die and not getting a 5 ✓

c) You catch the bus home

You don't catch the bus home ✓

11) [1, 1 = 2 marks]

There are 10 counters in a bag. The counters are numbered 1, 2, 3, 4, 5, 6, 7, 8, 9, 10.

If one counter is drawn out at random, what is the probability that it is a counter:

a) with an odd number

$$\frac{5}{10} = \frac{1}{2} \checkmark$$

b) that is greater than 6

$$\frac{4}{10} = \frac{2}{5} \checkmark$$

12) [2 mark]

The probability of selecting a blue marble from a bag of marbles is 0.3.

a) Find $P(\text{Not blue})$

$$1 - 0.3 = 0.7 \quad \checkmark$$

b) How many marbles could there be in the bag

10 marbles $\checkmark$ Any number that $P(\text{blue}) = 0.3$
gives a whole number of blue marbles

13) [1, 1 = 2 marks]

If we write the letters from the word MATHEMATICS on separate pieces of paper and place them in a bag, the probability of selecting an M is: $P(M) = \frac{2}{11}$.

a) Write down a word which has $P(D) = \frac{1}{4}$.

Date $\checkmark$ Any word

b) Write down a word which has $P(\text{vowel}) > P(\text{consonant})$.

heave $\checkmark$ Any word